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Subsurface Mapping and Characterization of Complex Sand Bodies Using Advanced Logging Technology and Innovative Solutions

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Abstract

In complex sand environments characterized by severe channeling discontinuities, conventional logging, coring, and traditional Geo-steering techniques struggle to provide accurate reservoir navigation and complex sand channels mapping, leading to reduced net-to-gross ratios and increased wellbore delivery times. The paper presents a case study where sand complexity challenges were overcome by developing and implementing a novel reservoir navigation method utilizing most advanced technology to improve geo-steering decisions and maximize well productivity at lower cost.

Advanced Electro-Magnetic Ultra Deep Azimuthal Resistivity (UDAR) technology was employed to achieve precise formation mapping. The tool's real-time ultra-high-definition (UHD) inversion and multi-frequency capabilities allowed simultaneous optimization of resolution and depth of detection (DOD). This technology enabled the team to map the complex sand structure within 100 feet radius of investigation. This approach allows for proactive adjustments to maintain the wellbore within the targeted sand layer, even in areas with severe lateral thickness variations and thin resistive sands. This data can be integrated into the dynamic reservoir simulation model to better map the sand channeling. By leveraging a combination of inversion-based analysis and advanced algorithms, we generate high-resolution images of the subsurface environment, enabling more proactive geo-steering decisions.

By applying this novel reservoir navigation technique, we successfully placed the wellbore through complex thin sand channels with varying resistivity areas, achieving significant improvements in net-to-gross ratio and reducing wellbore delivery time. Real-time mapping of reservoir boundaries enabled optimal well placement, resulting in enhanced productivity and improved returns on investment.

Introduction

Effective reservoir management and precise well placement are very important when people want to get oil and gas out of the ground, especially in places under the sea near Saudi Arabia. These places, called offshore fields, can be very tricky because the underground layers there are complicated. Getting oil out safely and efficiently helps produce more energy and also makes more money.

Sometimes, under the ground, there are rocks that look very similar to each other, making it hard to see clearly. These special rocks have very low differences in their electrical properties, which scientists usually rely on to identify oil or water. Also, underground layers often include thin, small pockets of sand that are not connected together, rocks that suddenly tilt in different directions, and structures that scientists are unsure about because they can't always see clearly. These issues make it very difficult for scientists to know exactly where to drill.

Because of these underground complications, it is easy for drills to go off track and miss the best spots to get oil or gas. When this happens, less oil and gas come out, and sometimes water flows in instead, which is not helpful. This can greatly reduce the amount of oil the field can produce over time and makes getting the energy out harder and less profitable.

The usual tools used to help guide the drilling process are called logging-while-drilling (LWD). These traditional tools try to look underground as they drill to find the best path. However, they can't see very far or very clearly, especially when the layers are thin, or when the rocks change quickly from one type to another. These thin layers and sudden changes in rock types make it even harder for the usual tools to understand what's happening underground. Also, when there is water deep underground, it can confuse these tools even more, increasing the chances of making mistakes during drilling.

To solve these problems, scientists needed to invent a better, more powerful tool. This new tool is called Advanced Ultra Deep Azimuthal Resistivity, or UDAR. UDAR was used in a special well named "Well X" in one of the major offshore oil fields in the region. UDAR is much better than the older tools because it uses eight different frequencies to explore underground, allowing it to see more clearly and further ahead. It also has special computer algorithms called ultra-high-definition (UHD) inversion, which can analyze the underground data more accurately and show a very detailed picture of the underground rock layers.

With these improvements, UDAR can map even very small and difficult-to-see underground features, helping scientists and engineers drill exactly in the right spot. By doing this, they get more oil out safely, even when the underground is complicated. This advanced technology greatly helps in improving how well scientists understand the reservoirs, ultimately leading to better decisions, higher oil production, and safer drilling practices.

Geological Background

The offshore field discussed in this study is one of the largest places under the sea in the region where scientists find oil and gas. This area is very special because the layers underground were formed from tiny pieces of rocks, sand, and clay that settled on the bottom of ancient oceans and rivers. These pieces slowly built up, layer by layer, forming complicated structures underground. Within these layers, the main areas scientists look for oil are thin pockets of sand, which sit above a large sandstone layer. These sand pockets are very important because they usually contain oil, but they are tricky because they aren't all connected and are mixed up with layers of clay-like rock called shale. This makes it difficult to know exactly where the oil is.

The sand pockets scientists look for are usually very thin, and they often disappear quickly or become extremely thin, making them hard to find and follow underground. Another tricky thing is that these sands can change a lot in their ability to hold electricity, called resistivity. In some places filled with water, the resistivity is very low, meaning it does not hold electricity well. But in other places filled with oil, the resistivity can be very high, meaning it holds electricity much better.

In addition to these tricky sand layers, the underground areas often have structures that are very complicated. The rocks can suddenly change their angle or direction, making it even harder for scientists to see clearly what's happening underground. There can also be tiny faults or cracks that are too small to be easily seen, even with special equipment. These small but complicated features add more confusion,

making it harder to understand the exact position of underground layers, and making it challenging to know exactly where to drill.

Challenges and Objectives of Well

Primary Challenges

When drilling Well X, scientists encountered several significant challenges. The underground reservoir exhibited considerable heterogeneity, with rapid lateral and vertical variations that complicated understanding. The stratigraphy could abruptly transition from sand to shale or clay, further adding to the complexity. Additionally, the rocks displayed low electrical contrast, making it difficult for conventional tools to distinguish between oil and water.

Another major challenge was the presence of water beneath the oil reservoirs. This water posed a risk of infiltrating the drilled hole if not managed carefully, potentially flooding the oil and hindering its extraction. Furthermore, the sands were characterized by their thin, small, and disconnected nature, which increased uncertainty and made it challenging for scientists and drillers to accurately navigate the subsurface.

To address the challenges encountered while drilling Well X, the contracting company set clear objectives. The plan was to drill a horizontal well section with a diameter of 8.5 inches and a length of 4,000 feet, targeting specific thin sand pockets within the subsurface. This precision ensured the well was positioned directly within the oil-rich zones.

Additionally, maintaining an optimal distance from the underlying water zones was crucial to prevent premature water intrusion into the oil reservoirs. Another key goal was to accurately map and identify the boundaries of the oil reservoirs in real-time during drilling. This enabled the team to stay precisely on course and maximize oil extraction from the reservoir.

Fit-for-Purpose Technology

Given the geological complexities of the reservoir, scientists and engineers selected the advanced Ultra-Deep Azimuthal Resistivity (UDAR) technology to address the limitations of traditional resistivity-based tools. UDAR technology incorporates multi-frequency operations ranging from very low frequencies (1 KHz) to very high frequencies (2 MHz). This broad frequency range allows for detailed geological characterizations across varying depths, from immediate proximity to the borehole to deeper geological environments beyond the capabilities of conventional tools.

Additionally, UDAR employs ultra-high-definition (UHD) inversion algorithms, enhancing subsurface imaging resolution and the accuracy of geological interpretations. By utilizing these advanced inversion algorithms, the UDAR system provides detailed and reliable real-time mapping of reservoir boundaries and geological structures, even in heterogeneous and resistivity-challenged environments. Consequently, this technology improves the accuracy of well placement, reduces uncertainties, and increases hydrocarbon recovery efficiency.

Advanced UDAR Tool Description

The advanced UDAR tool deployed in Well X is engineered with numerous advanced capabilities, specifically designed for detailed and precise subsurface investigation. Its multi-frequency functionality, operating across a comprehensive frequency range from as low as 1 KHz to as high as 2 MHz, allows for effective analysis of geological formations at various depths. The tool's advanced antenna system consists of three orthogonally oriented antennas, each precisely positioned to generate full tensor resistivity measurements, thereby significantly enhancing the clarity and precision of subsurface imaging due to higher signal-to-noise output, pure X, Y and Z magnetic field components, which greatly improves signal quality by effectively minimizing environmental and instrumental noise, facilitating more accurate real-time data

acquisition. The UDAR system provides a complete 360-degree azimuthal imaging capability, enabling precise detection and delineation of reservoir boundaries and structural features even in highly complex geological settings.

Operationally, the advanced UDAR tool has been meticulously engineered for borehole compatibility, designed to fit seamlessly within an 8½-9 7/8 inch borehole sizes and measuring approximately 13.8 feet in length. It is adaptable to both conductive and non-conductive drilling fluids and maintains optimal functionality under challenging drilling environments, including conditions with high fluid flow rates and elevated temperatures reaching up to 302°F (150°C). Its robust construction and flexible design ensure high reliability and consistent performance, even under the demanding conditions characteristic of offshore drilling operations. Figure 1 below presents a three module configuration, including all frequencies and three measurements components, XZ, ZX and ZZ.

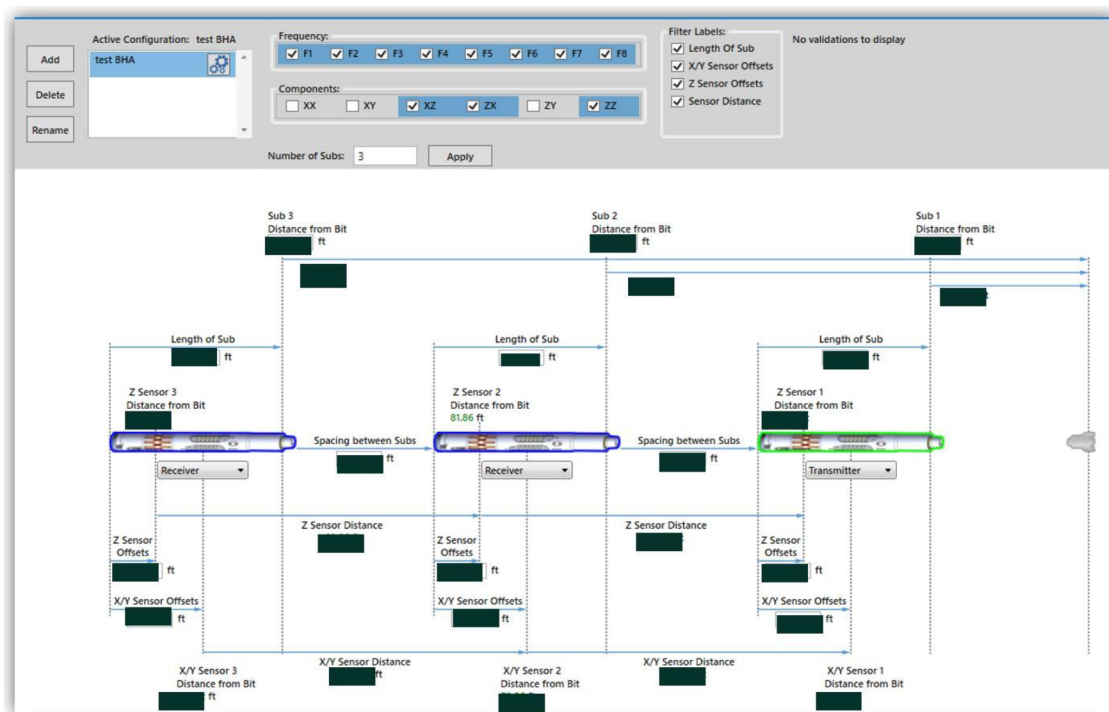


Figure 1—BHA Schematic & Sensor offset

In addition, advanced UDAR tool implements the new transceiver concept, which permits all modules to be capable of transmit and receive signal. This innovation add flexibility to BHA and allows different spacings of same measurements from same geological formations within depth of detection of the tool. Thus, it has an operational advantage in case of any antenna failure, since each module can transmit or receive any of the frequencies of components mentioned before.

Inversion Workflow

The advanced UDAR system employs a sophisticated hybrid inversion workflow designed specifically to maximize geological interpretation resolution and accuracy, effectively addressing complex reservoir conditions. The workflow encompasses several advanced methodologies, integrating both parametric and artificial intelligence-based inversion algorithms.

Initially, Parameter Domain Scanning is conducted systematically to evaluate multiple geological scenarios, refining the geological models through comprehensive resistivity responses (Fang et al., 2005; Helgesen et al., 2005). This rigorous scanning approach thoroughly examines various geological

parameters and scenarios, significantly improving model calibration and ensuring the accuracy of resistivity measurements interpretation. The selection of specific subsets of data for either proactive reservoir navigation or comprehensive geo-mapping objectives is guided by the specific reservoir challenges and geological intricacies encountered (Antonsen et al., 2018).

Forward Modeling follows as an essential step, where simulated resistivity responses derived from predefined geological models and parameter settings are compared against actual measured resistivity data (Fares, 2023). This validation step is crucial in assessing the fidelity and reliability of inversion results, enhancing confidence in real-time geological interpretations. The modeling ensures the accurate depiction of both subtle and significant geological features such as thin layer pinch-outs, fluid boundaries, and structural variations (Antonsen et al., 2018).

In scenarios where conventional one-dimensional (1D) or even two-dimensional (2D) inversion methods are insufficient, particularly in environments exhibiting significant lateral heterogeneities such as faults and angular unconformities, hybrid 2D inversion techniques leveraging Artificial Neural Networks (ANN) are employed (Kushnir et al., 2018; Astrakova et al., 2021). ANN-based inversions significantly accelerate the interpretation process, providing ultra-high-definition (UHD) visualization of subsurface resistivity structures and enhancing the characterization of complex geological features in real-time (Kushnir et al., 2018).

Finally, High-Performance Computation is integrated into this workflow, leveraging robust computational clusters to handle extensive data processing requirements efficiently. This computational capability is critical for real-time decision-making during drilling operations, supporting rapid analysis of extensive data sets generated by the UDAR tool (Hartmann et al., 2014). The high-performance computational resources enable immediate processing and visualization of UHD inversion results, facilitating swift and informed geosteering decisions that significantly enhance operational efficiency and reservoir navigation accuracy (Ferreira et al., 2024).

The integration of these detailed inversion outputs into reservoir simulation models significantly enhances the accuracy of reservoir characterization. By providing precise and reliable data on reservoir structure, fluid distribution, and geological heterogeneity, the models become better calibrated and thus deliver more reliable forecasts. This improved predictive capability allows for optimized strategic planning, reducing uncertainty and enhancing overall field development decisions, ultimately maximizing economic returns and reservoir productivity.

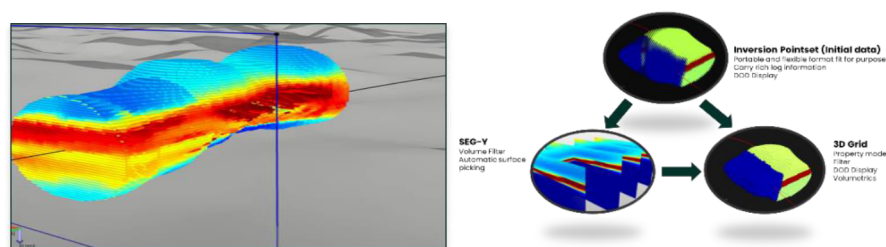


Figure 2—Example of UHD inversion technology

Application of UDAR Technology

Pre-Well Planning and Feasibility Study

A comprehensive pre-well feasibility study was meticulously conducted by integrating extensive offset well data to construct an accurate and predictive inversion model. This advanced modeling approach utilized historical data from nearby wells to rigorously evaluate potential geological variations and resistivity characteristics expected in the targeted reservoir interval. By effectively incorporating this detailed offset

well information, the predictive inversion model demonstrated robust capabilities in accurately delineating resistivity variations across the vertical depth range of 93-108 ft True Vertical Depth (TVD).

The model's ability to precisely predict these resistivity changes provided exceptionally high confidence levels for optimal well placement strategies, significantly reducing geological uncertainties during drilling operations. Additionally, this pre-well feasibility study enabled early identification and management of potential geological complexities, including sudden facies changes, reservoir boundary variations, and the presence of conductive water zones. This comprehensive approach ensured precise planning and efficient execution of the drilling strategy, ultimately leading to optimized hydrocarbon recovery.

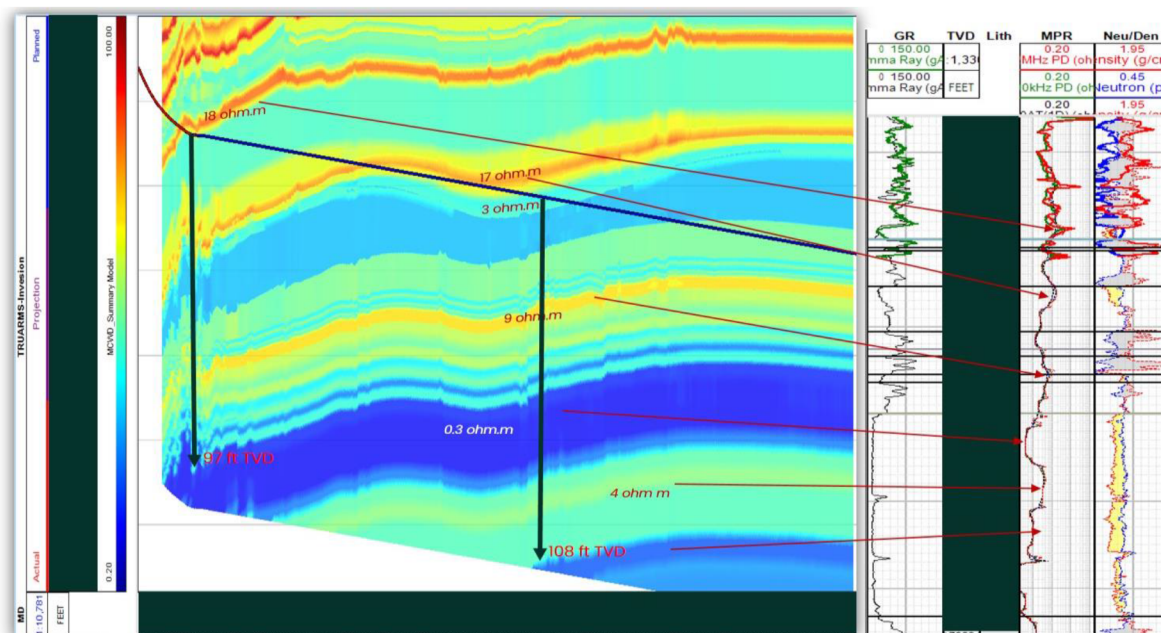


Figure 3—Pre-well modeling and feasibility study

Real-Time Monitoring

During the actual drilling operations, the advanced UDAR system demonstrated outstanding real-time subsurface mapping capabilities. Its precise and high-resolution data acquisition and processing allowed for immediate identification and detailed characterization of critical geological features. Specifically, the UDAR system accurately distinguished and delineated between 13 to 16 distinct thin sand layers, achieving precise measurement of individual layer thicknesses down to subtle and intricate pinch-out features. This capability significantly enhanced the accuracy of reservoir characterization and allowed immediate adjustments to drilling trajectories.

Furthermore, the UDAR system clearly mapped conductive fluid boundaries, specifically identifying oil-water contacts at approximately 80 to 90 ft TVD beneath the wellbore. These delineations provided crucial real-time insights into fluid distribution, significantly mitigating risks associated with premature water breakthrough and enhancing overall reservoir management strategies.

Notably, a distinct U-shaped geological feature was precisely mapped midway through the well trajectory. This previously ambiguous geological structure became clearly defined, substantially improving local geological understanding and aiding in strategic well navigation decisions.

The provision of such detailed, high-resolution, real-time mapping data significantly improved the effectiveness of geosteering operations. Immediate and informed adjustments to drilling paths ensured consistent, optimal reservoir contact throughout the drilling process, maximizing hydrocarbon recovery and operational efficiency.

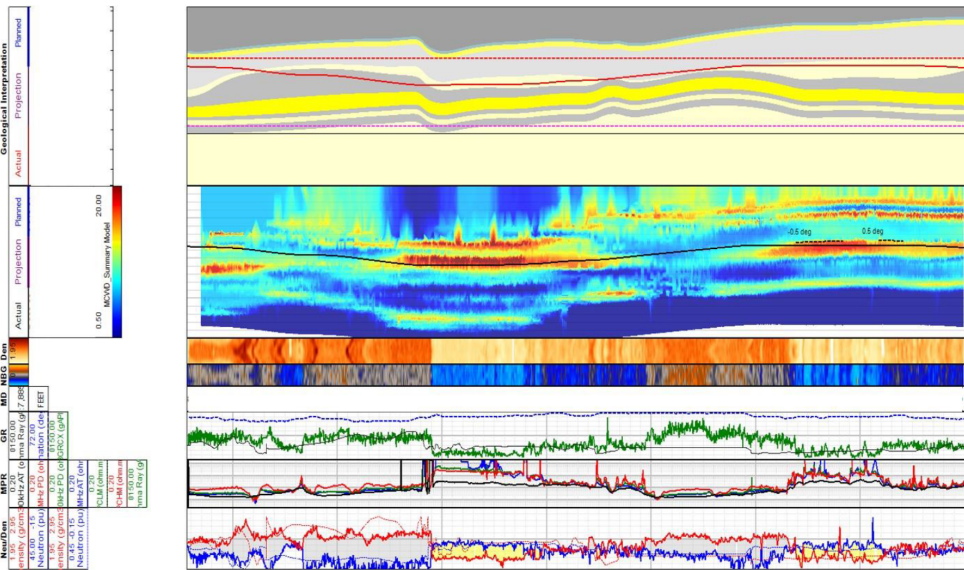


Figure 4—Real-Time Results and UHD Inversion Mapping

Post-Well Analysis

Following the completion of drilling operations, a comprehensive post-well analysis was conducted, extensively utilizing the outputs from UHD inversion data. This rigorous analytical approach significantly enhanced overall reservoir characterization by meticulously validating pre-well predictive models and illuminating previously unresolved geological complexities. High-resolution inversion results enabled detailed characterization of subtle geological features and fluid distribution patterns, substantially refining the understanding of reservoir architecture and heterogeneity.

The UHD inversion outputs were seamlessly integrated into advanced reservoir simulation models, improving the accuracy and reliability of predictive simulations. This integration provided robust inputs that significantly refined future drilling strategies and enhanced reservoir development planning. By accurately characterizing reservoir properties and geological features, this extensive post-well analysis substantially improved forecasting capabilities, ensuring informed decision-making for subsequent well placements and overall field management strategies.

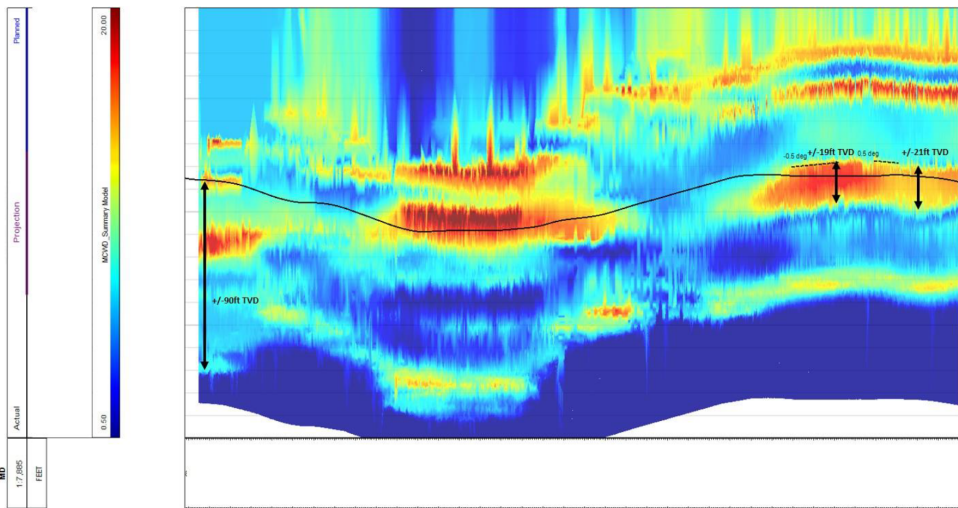


Figure 5—Detailed UHD Inversion Result showing real-time mapping

Discussion

The deployment of advanced UDAR technology significantly improved geological understanding and reservoir characterization within the complex offshore field under study. Compared to traditional reservoir navigation approaches, the enhanced depth of detection and superior resolution capabilities of UDAR technology provided unprecedented insights into subtle geological variations, reservoir continuity, and complex fluid distributions. This allowed for the clear identification of fine-scale features such as thin sand layers, abrupt facies transitions, and conductive fluid interfaces.

The sophisticated inversion workflow associated with UDAR technology substantially reduced operational uncertainties, facilitating more precise well placement and improved reservoir contact. Consequently, this contributed directly to enhanced economic outcomes by increasing hydrocarbon recovery and minimizing drilling risks associated with uncertain geological scenarios. Additionally, the seamless integration of UDAR data into subsurface modeling software streamlined the interpretation process, greatly simplifying post-well analysis and significantly enhancing future planning and reservoir development strategies.

Conclusions

The deployment of advanced UDAR technology effectively addressed the significant geological complexities encountered in the offshore field, resulting in substantially improved reservoir navigation accuracy. This advancement directly enhanced the potential for increased hydrocarbon recovery by providing precise and continuous reservoir contact throughout the drilling process.

The real-time UHD inversion capability of the UDAR system successfully and accurately delineated critical fluid boundaries, particularly oil-water contacts (OWC), and subtle geological features at substantial distances, reaching up to 90 feet away from the wellbore. This precise mapping significantly improved geological understanding and enabled proactive reservoir management.

Real-time reservoir navigation decisions driven by high-resolution inversion data facilitated substantial improvements in net-to-gross ratios, effectively increasing the economic viability of the drilled well. Additionally, these informed, data-driven decisions substantially minimized operational risks, mitigating the potential for costly drilling complications and unplanned water influx.

Post-well integration of comprehensive inversion results markedly enhanced the overall characterization of reservoir properties and geological architecture. This improved understanding positively influenced strategic planning, enabling optimized and informed future development decisions and reservoir management strategies.

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